

REVISED TABLES TO ACCOMPANY "STARNAP"

AVERAGED PLANETARY ELEMENTS, 1 JANUARY 1960

	SEMINAJOR	PERIOD	ECCENT'Y	INCLINAT'N	ASCENDING	PERIANGLE	ANOMALY	SECULAR ASCENDING	SECULAR PERIANGLE	JULIAN DATE
MERCURY	0.387099	0.24085	.205627	7.00399	47.85714	76.83309	145.78856	0.0000324444198033	0.000042559243364	2436935.5
VENUS	0.723332	0.61521	.006793	3.39423	7.63197	131.00831	43.286	0.000024641163063	0.000038505619934	2436935.5
EARTH	1	1.00004	.016726	0	0	102.25253	-2.09438	0	0.000047000736953	2436935.5
MARS	1.523691	1.88089	.093368	1.84991	49.24903	335.32269	-76.5554	0.000021188358115	0.000050392699523	2436935.5
JUPITER	5.202803	11.86223	.048435	1.30536	100.04444	13.67823	246.15289	0.00002768328196	0.000044110724001	2436935.5
SATURN	9.538843	29.45772	.055682	2.48991	113.30747	92.26447	188.40688	0.000023880633339	0.000053617345554	2436935.5
URANUS	19.181951	84.01331	.047209	0.77306	73.7963	170.01083	-28.70587	0.000013689535035	0.0000044110724001	2436935.5
NEPTUNE	30.057779	164.79345	.008575	1.77375	131.3398	44.27395	172.66695	0.000030040924104	0.00001825271338	2436935.5
PLUTO	39.43871	247.686	.250236	17.1699	109.88562	224.16024	-42.51392	0.000038026486208	0.000038026486208	2436935.5

PLANETARY ELEMENTS, 4 APRIL 1978

	SEMINAJOR	PERIOD	ECCENT'Y	INCLINAT'N	ASCENDING	PERIANGLE	ANOMALY	SECULAR ASCENDING	SECULAR PERIANGLE	JULIAN DATE
MERCURY	0.387099	0.24085	.20563	7.00433	48.07347	77.11704	63.2437	0.0000324444198033	0.000042559243364	2443600.5
VENUS	0.723332	0.61521	.006783	3.39442	76.48402	131.26501	282.2849	0.000024641163063	0.000038505619934	2443600.5
EARTH	1	1.00004	.016718	0	0	102.56635	87.4242	0	0.000047000736953	2443600.5
MARS	1.523691	1.88089	.093385	1.84981	49.38973	335.65866	176.3051	0.000021188358115	0.000050392699523	2443600.5
JUPITER	5.202432	11.86223	.0479396	1.30554	100.2155	14.1085	79.9636	0.00002768328196	0.000044110724001	2443600.5
SATURN	9.578194	29.45772	.0575851	2.48668	113.491	94.4031	49.1255	0.000023880633339	0.000053617345554	2443600.5
URANUS	19.17824	84.01529	.0480776	0.77141	74.0002	168.8653	50.6774	0.000013689535035	0.000044110724001	2443600.5
NEPTUNE	29.96499	164.78829	.0111899	1.77207	131.5474	59.5713	197.3397	0.000030040924104	0.00001825271338	2443600.5
PLUTO	39.54322	248.4302	.2493373	17.13713	109.8868	223.1483	343.6598	0.000038026486208	0.000038026486208	2443600.5

PLANETARY ELEMENTS, 7 MARCH 1973

	SEMINAJOR	PERIOD	ECCENT'Y	INCLINAT'N	ASCENDING	PERIANGLE	ANOMALY	SECULAR ASCENDING	SECULAR PERIANGLE	JULIAN DATE
MERCURY	0.387	0.24085	.2056	7.00389	48.02154	77.04888	289.655	0.0000324444198033	0.000042559243364	2441721.5
VENUS	0.723	0.61521	.0068	3.394167	76.2275	130.8675	150.2801	0.000024641163063	0.000038505619934	2441721.5
EARTH	1	1.00004	.0167	0	0	102.08056	34.4957	0	0.000047000736953	2441721.5
MARS	1.524	1.88089	.0934	1.84981	49.17194	335.13861	271.146	0.000021188358115	0.000050392699523	2441721.5
JUPITER	5.203	11.86223	.0484	1.305833	99.94333	13.51722	283.9167	0.00002768328196	0.000044110724001	2441721.5
SATURN	9.539	29.45772	.0557	2.490278	113.22	92.06833	348.2963	0.000023880633339	0.000053617345554	2441721.5
URANUS	19.191	84.01529	.0472	0.773056	73.74	169.85139	25.4394	0.000013689535035	0.000044110724001	2441721.5
NEPTUNE	30.071	164.78829	.0086	1.774444	131.22833	44.15861	224.9265	0.000030040924104	0.00001825271338	2441721.5
PLUTO	39.518	248.4302	.2486	17.143889	109.65111	233.19278	335.6904	0.000038026486208	0.000038026486208	2441721.5

LUNAR ELEMENTS, 9 DECEMBER 1899

	SEMINAJOR	PERIOD	ECCENT'Y	INCLINAT'N	ASCENDING	PERIANGLE	ANOMALY	SECULAR ASCENDING	SECULAR PERIANGLE	JULIAN DATE
MOON	1	0.07544	.0549005	5.143412	260.38369	331.80423	0	-0.052953922	0.1114040803	2414997.331

* NOTE REVISION TO LINE [7]

^ ENTRY:W

[1] W+21

[2] STATEDAYNO+GETDATE

[3] STATEDIME+GETTIME

[4] LAT+GETLAT

[5] LONG+GETLONG

[6] TIME+LONG TIMEADJUST STATEDIME

[7] DATE+STATEDDAYNO+0.5*(TIME+24)-LONG+360

^

* NOT USED IN STARNAP, BUT INCORRECTLY STATED ON P. 12

^ Z+PERIODE

[1] Z+ANOMALYDATE-(ANOMALY+360)*PERIOD+TROPYR

^